WORKSHEET 1.11

Unit 1 Test: Answer Key

Unit 1: Limits and Continuity · 20 points

Answers

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	B	D	B	B	A	C	A	C	A
Question	11	12	13	14	15	16	17	18	19	20
Answer	D	B	C	D	A	C	B	D	A	C

Worked solutions

1. $\dfrac{(x-2)(x+2)}{(x-2)(x+3)} = \dfrac{x+2}{x+3} \to \dfrac{4}{5}$. **(A)** Lesson 1.5

2. Both branches approach the open circle at height 0; the filled dot at height 2 is $q(-2)$. **(B)** Lesson 1.2

3. The left-hand limit is 3 and the right-hand limit is 4, so the limit does not exist. **(D)** Lesson 1.2

4. $\dfrac{3}{4} \cdot \dfrac{\sin(3x)}{3x} \to \dfrac{3}{4}$. **(B)** Lesson 1.6

5. Equal degrees: $\dfrac{6}{2} = 3$. **(B)** Lesson 1.9

6. $\sqrt{16x^2+3} = -x\sqrt{16+\frac{3}{x^2}}$ for $x<0$, so the limit is $\dfrac{-4}{2} = -2$. **(A)** Lesson 1.9

7. $1+k = 3+2k$ gives $k=-2$. **(C)** Lesson 1.8

8. $\dfrac{(x-5)(x+5)}{(x-5)(x+1)}$: the factor cancels, and the limit $\frac{10}{6}$ exists. **(A)** Lesson 1.7

9. The sign changes on $[0,2]$, $[2,5]$ and $[5,8]$: at least 3 zeros, by the Intermediate Value Theorem. **(C)** Lesson 1.10

10. $\dfrac{\sqrt{x}-3}{(\sqrt{x}-3)(\sqrt{x}+3)} \to \dfrac{1}{6}$. **(A)** Lesson 1.5

11. $\dfrac{3(2)-(-1)^2}{2+(-1)} = \dfrac{5}{1} = 5$. **(D)** Lesson 1.4

12. Both bounds tend to 4, so by the squeeze theorem the limit is 4. **(B)** Lesson 1.6

13. Numerator near 4; denominator small and negative. **(C)** Lesson 1.9

14. As $x \to 1^+$, $q(x) = 5-x \to 4$ through values less than 4, and $\lim\limits_{u\to 4^-} q(u) = 5-4 = 1$. **(D)** Lesson 1.2

15. $\dfrac{x-1}{(x-1)(x+1)} = \dfrac{1}{x+1}$ for $x \neq 1$: a hole at $x=1$, an asymptote at $x=-1$. **(A)** Lesson 1.7

16. A table at $x = \pm 0.001$ gives 2.002 and 1.998. **(C)** Lesson 1.3

17. A table at $x = \pm 0.01$ gives 4.4997. **(B)** Lesson 1.3

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18. The entries on both sides approach 5; $g(2) = 1$ is the function value. **(D)** Lesson 1.3

19. A table at $x = 1 \pm 0.001$ gives 0.9995 and 1.0005. **(A)** Lesson 1.3

20. h is continuous and $h(1) < 0 < h(4)$, so the Intermediate Value Theorem gives a zero. It does not give uniqueness. **(C)** Lesson 1.10

Using your score

One point per question. 17 or more: ready to move on. 13 to 16: rework the lessons named beside the questions you missed, then try the online practice for those lessons. 12 or fewer: review the unit summary pages before continuing to Unit 2.

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